MIL-A-81423(AS)
Amendment -2

12 October 1972
Superseding
Amendment -1
28 September 1967

MILITARY SPECIFICATION

ATTITUDE HEADING REFERENCE SYSTEM AN/ASN-73

This amendment forms a part of Military Specification MIL-A-81423(AS) dated 1 October 1966 and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 1

1.2: Delete "CN-105A/A" and "C-2984/AJB-3A" and substitute

"...CN-1054A/A..." and "C-2984/AJB..."

Page 2

2.1 Add the following under military specifications:

| | |
|---|---|
| MIL-L-6085 | Lubricating Oil, Instrument, Aircraft, Low Volatility |
| MIL-G-81322 | Grease, Aircraft, General Purpose Wide Temperature Range |
| MIL-B-81744 | Barrier Coating Solution and Solvent |

Page 3

Add the following under military standards:

| | |
|---|---|
| MIL-STD-1334 | Process for Barrier Coating of Anti-Friction Bearings |

Add the following under Naval Air Systems Command:

| | |
|---|---|
| AR-34 | Failure Classification for Reliability Testing, General Requirements for |
| AS-2844 | Lubricant, Instrument Ball Bearing, High Flash Point |

FSC 6610

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Page 7

3.3.2.4: Delete "MS17323" and substitute

"MS17322-8A"

Page 9

3.3.10.1: Delete and substitute

"3.3.10.1 Vibration - The equipment, less the Compass Controller, shall operate satisfactorily when subjected to the vibration requirements of Curve IV and II of Specification MIL-E-5400. The Compass Controller shall operate satisfactorily when subjected to the vibration requirements of Curve III of Specification MIL-E-5400 except that the maximum acceleration shall be limited to two (2) G's."

3.3.11, lines 3 and 4: Delete "...3.5 minutes .... and 5 minutes..." and substitute

"...5 minutes... and 15 minutes..."

Page 10

3.3.13.2, line 10: Delete "such that it will pass the case lead test." and substitute

"such that it will pass the case leak test."

Add the following new paragraphs:

* 3.3.16 Lubrication -

* 3.3.16.1 Rotor Bearing Lubrication - The rotor bearings shall be lubricated with grease conforming to MIL-G-81322 or MINIPURE grease with particle size not exceeding 50 microns.

* 3.3.16.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall be AS-2844 in accordance with the requirements of MIL-L-6085.

* 3.3.16.3 Barrier Film Bearing Coating - All bearings shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings, including chamfered races or corners, and including interfacing parts where applicable, shall be barrier coated.



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* 3.3.16.4 Ball Bearings - Ball bearings shall be ABEC-5 grade or better. Teflon separators shall be used on gimbal bearings where the bearing design permits. Spin rotor bearings shall have porous separators impregnated with lubricant AS-2844 and shall be ABEC-7 or better.

* 3.3.16.5 Ball Bearing Pre-Loading - Means shall be provided for pre-loading and alignment without causing eccentric loading unless otherwise authorized.

Page 11

3.4.3, line 3: Delete "the statistical accuracies of output signals" and substitute

"the statistical one sigma accuracies of output signals"

Page 18

3.5.1.1.5, paragraph 3, line 5: Delete "Table II" and substitute

"Table I"

3.5.1.1.5.1.1: Delete and substitute

"3.5.1.1.5.1.1 Externally Excited Pitch Synchro - Terminals c and K, Table I, shall be connected to the power lead terminal A, Table II, and terminal L of Table I shall be connected to the power lead terminal D, Table II. The voltage across terminals a and b, and the voltage across terminals L and a or b of Table I shall be measured with the gimbals at zero displacement. The voltage across L and a or b of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals a and c, and b and c, of Table I shall be measured as the reference assembly is rotated in climb about the pitch gimbal. The voltage across terminals a and c, Table I, shall increase before decreasing and the voltage across the terminals b and c, Table I, shall decrease before increasing."

Page 23

3.5.1.1.5.1.3: Delete and substitute

"3.5.1.1.5.1.3 Inner Roll Synchro - Terminal j of Table I shall be connected to terminal A of Table II. The voltage across g and h of Table I and the voltage across terminal D of Table II and terminal g or h of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals g and h shall be at the minimum and the voltage across terminal D of Table II and terminals g or h of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals g and j and across h and j of Table I



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3.5.1.1.5.1.3 (Continued) shall be measured as the first gimbal (inner roll) is held and the reference assembly with the outer roll gimbal is rotated in right roll. The voltage across terminals g and j of Table I shall increase before decreasing and the voltage across terminals h and j shall decrease before increasing."

3.5.1.1.5.1.4: Delete and substitute

"3.5.1.1.5.1.4 Internally Excited Pitch Synchro - Terminal f of Table I shall be connected to terminal A of Table II. The voltage across d and e of Table I and the voltage across terminal D of Table II and terminal d or e of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals d and e shall be at the minimum and the voltage across terminal D of Table II and terminal d or e of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals d and f and across e and f of Table I shall be measured as the second gimbal (pitch) is held and the reference assembly with theouter roll gimbal is rotated in climb. The voltage across terminals d and f of Table I shall increase before decreasing and the voltage across terminals e and f shall decrease before increasing."

Page 24

3.5.1.1.5.1.5, line 7: Delete "Table I" and substitute

"Table II"

Page 25

3.5.1.1.9, lines 4 and 7: Delete "±90" and substitute

"+90"

Page 26

3.5.1.2.4.1, line 7: Add at the end of the sentence

"The synchro shall not exceed a maximum transmission error of 8 minutes of arc."

Page 27

3.5.1.2.4.3, lines 9 and 18: Delete "Table II" and substitute

"Table I"

3.5.1.2.4.3, line 10: Delete "Table I" and substitute

"Table II"



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3.5.1.2.4.3.1, line 2: Delete "Table I" and substitute

"Table II"

Page 29

3.5.1.3, line 3: Add page "30."

Page 40

Table VIII: Delete "To mate with MS3116P-10-6S or equal (5 req'd)" and substitute

"To mate with MS3116P-10-6S or equal (4 req'd)"

Table VIII: Delete

| Terminal | Function |
|---|---|
| D | CX Excitation C |
| E | CX Excitation H |

and substitute

| Terminal | Function |
|---|---|
| D | CX Excitation H |
| E | CX Excitation C |

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Table X: Add

| Terminal | Function |
|---|---|
| G | +28 vdc rectified |
| J | Spare |
| K | Spare |

Page 43

Table XI: Delete "Pre-Amplifier-Limited Output" and substitute

"Pre-Amplifier-Limiter Output"

Table XI: Delete "Slaving and Correction Motor" and substitute

"Slaving and Correction Velocity Generator"



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Page 45

3.5.3.4, line 5: Delete "Compensator shall increase (+) when the" and substitute

"Compensator shall increase (+) when the"

3.5.3.5, line 8: Delete "Items a, b, d, and e shall be" and substitute

"Items a, b, and d shall be"

Page 46

3.5.3.7, paragraph 2, line 1: Delete "With the CX (J908) at" and substitute

"With the CX (J906) at"

3.5.3.7, paragraph 2, lines 2 and 3: Delete "CX's (J906, J907 and J909)" and substitute

"CX's (J908, J907 and J909)"

Page 47

3.5.3.9.4: Delete and Substitute

"3.5.3.9.4 Synchronization Indicator - The null current through the 1100 ohm resistor shall not exceed 8 microamperes. The current through the 1100 ohm resistor, throughout a 5 degree displacement of the magnetic comparison transmitter shall be 10 ± 4 microamperes per degree"

3.5.3.10: Delete "(2) 23.5V, 400 cps for excitation of the type MX-1 magnetic compass transmitter." and substitute

"(2) 23.5V 400 cps for excitation of the type ML-1 magnetic compass transmitter."

Page 48

3.5.4: Delete "Controller-Compass System C-2984/AJB-3A" and substitute

"Controller-Compass System C-2984/AJB"

Page 49

Figure 5: Delete title and substitute

"Controller Compass System C-2984/AJB"



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4.4.1.3 Delete (Under Vibration) "Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and minimum of 3G's." and substitute "any selected frequency within the range of 50 to 60 cps (excluding resonant points) and a minimum of 3G's."

Page 60

4.4.3 Add the following to end of paragraph:

* Classification of a failure shall be in accordance with MIL-STD-781 and AR-34.

Add the following paragraph:

* 4.4.3.1.1 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance Testing shall be conducted through a production burn-in. Each equipment produced but not submitted to the Qualification Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be totaled and compared to the Accept/Reject Criteria of Test Plan II of MIL-STD-781. (Extend the lines as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. Delivery of equipment shall not be held up to await a decision under the test plan. In event an Accept Decision is reached, the test duration of each equipment shall be reduced to 100 operating hours. At any time that the current total of test hours and test failures plotted on Test Plan IV curves (extended) show a Reject Situation, the Procuring Activity shall be notified. The Procuring Activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision. For the Production Acceptance Phase, Test Level E shall be used, except that the temperature extremes shall be +55C and -44C, and vibration shall be 0.5G. Details of the burn-in tests shall be included in the test procedures.

Custodian:
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Preparing Activity:
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Project No. 6610-N065



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